

1) Given: $E = 19.1$ $\rho_{UN} = 14.3 \text{ g/cm}^3$ ~~$\theta = 587 \text{ b}$~~
 $M = 14$ ~~ϕ~~ $\phi = 4 \times 10^{12} \text{ n/cm}^2/\text{s}$

Find: • heat generation rate $\rho_{\text{UO}_2} = 10.97 \text{ g/cm}^3$

• enrichment of UO_2 to get same Q

Assume: $E = 200 \times 10^6 \text{ eV}$

Solve: $Q = E_f N_f \theta_f \phi$

$N_f = \frac{\rho_{UN}}{M_{UN}} (\text{ratio}) (\text{enrichment})$

UN
 $\begin{matrix} \nearrow \\ 235 \end{matrix}$
 $\begin{matrix} \searrow \\ 238 \end{matrix}$

~~$M_{UN} = 235.04$~~

$N_f = \frac{14.3 \text{ g/cm}^3 (6.022 \times 10^{23} \frac{\text{atom}}{\text{mol}})}{251.43 \text{ g/mol}} \left(\frac{1 \text{ atom U}}{1 \text{ molecule UN}} \right) (0.19)$

$N_f = 6.507 \times 10^{21} \text{ atom/cm}^3$

$M_{UN} = (0.19 \times 235) + (0.81 \times 238) + \frac{14}{N} = 251.43 \text{ g/mol}$

~~$M_{UN} = 235.04$~~

~~$M_{UN} = 251.43 \text{ g/mol}$~~

~~$N_f = 6.507 \times 10^{21} \text{ atom/cm}^3$~~

~~$\phi = 4 \times 10^{12} \text{ n/cm}^2/\text{s}$~~

$Q = (200 \times 10^6 \text{ eV}) \left(\frac{6.507 \times 10^{21} \text{ atom}}{\text{cm}^3} \right) (587 \times 10^{-24} \text{ cm}^2) \left(4 \times 10^{12} \frac{\text{n}}{\text{cm}^2/\text{s}} \right)$

$(1.602 \times 10^{-19} \frac{\text{J}}{\text{eV}}) = 489.56 \text{ W/cm}^3$

1 cont.)

n = enrichment

Crystal B.

enrich UO_2 to get $Q = 489.56 \text{ W/cm}^3$?

$$M = (n \times 235) + [(1-n)238] + (2 \times 16) =$$

$$M_{UO_2} = 235n + (1-n)238 + 32$$

$$N_f = \frac{\rho Na}{M_{UO_2}} \text{ ratio} \cdot n$$

$$N_f = \frac{(10.97 \text{ g/cm}^3) (6.022 \times 10^{23} \frac{\text{atoms}}{\text{mole}})}{[235n + 238(1-n) + 32]} \cdot \frac{1 \text{ U}}{1 UO_2} \cdot n$$

$$Q = E_f N_f Q_f \phi$$

$$489.56 \frac{\text{W}}{\text{cm}^3} = (200 \times 10^6 \text{ eV}) (1.602 \times 10^{-19} \frac{\text{J}}{\text{eV}}) (587 \times 10^{-24} \text{ cm}^2) (4 \times 10^{12} \text{ n/cm}^2 \cdot \text{s}) \cdot N_f$$

$$6.5075 \times 10^{21} = N_f$$

$$\frac{6.5075 \times 10^{21}}{6.51 \times 10^{21}} = \frac{(6.61 \times 10^{24})n}{235n + 238(1-n) + 32}$$

$$\frac{1.53 \times 10^{24}n + 1.55 \times 10^{24}(1-n) + 2.08 \times 10^{23}}{n} = 6.61 \times 10^{24}$$

1 cont. }

Chapter 10

$$1.53 \times 10^{24} + 1.55 \times 10^{24} \frac{(1-n)}{n} + \frac{2.08 \times 10^{23}}{n} = 6.6 \times 10^{24}$$

$$1.55 \times 10^{24} \frac{(1-n)}{n} + \frac{2.08 \times 10^{23}}{n} = 5.08 \times 10^{24}$$

$$1.55 \times 10^{24} (1-n) + 2.08 \times 10^{23} = 5.08 \times 10^{24}$$

ran out of
time

2) temp at radial midpoint of cladding
and at $r = 0.1 \text{ cm}$.

$\text{Gas} = \text{He}$

$$T_{oc} - T_{cool} = \frac{Q}{2h_{cool}} \quad R_f = \frac{LHR}{2\pi R_c h_{cool}}$$

~~Temperature~~

$$T_{oc} = T_{cool} + \frac{350 \text{ W/cm}^3 \cdot 0.45 \text{ cm}}{2 \cdot 1.5 \text{ W/cm}^2\text{K}} = 540 \text{ K} + 592.5 \text{ K}$$

$$T_{ic} = T_{oc} + \frac{350 \cdot 0.45}{2 \cdot 3} = 618.75 \text{ K}$$

$$h_{clad} = \frac{k}{t} = \frac{0.18 \text{ W/mK}}{0.06 \text{ cm}} = 3$$

$$\begin{array}{ccccccc} & & | & & | & & \\ \text{Gas} & & T_{0.45} & & T_s & & \\ & & | & & | & & \\ & & 618.75 & & 350 \times 0.45 & & \\ & & | & & | & & \\ & & T_{ic} & & T_{oc} & & \end{array}$$

$$T_s = T_{ic} + \frac{350 \times 0.45}{2 \cdot 0.32} = 864.8 \text{ K}$$

$$80 \times 10^{-6} = 0.00008 \text{ m} \left(\frac{10 \text{ cm}}{1 \text{ m}} \right)$$

$$k_{\text{gas}} = 16 \times 10^{-6} \quad (618.75 \text{ K})^{0.79} = 0.00257$$

$$h_{\text{gas}} = k/t = 0.00257 / 0.008 = 0.32$$

$$T_0 = 864.8K + \frac{350}{4 \cdot 0.04} 0.45^2 = 1307.8K$$

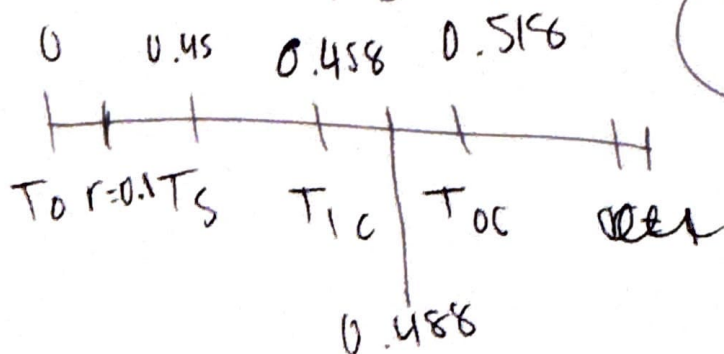
$$\begin{matrix} T_1 & T_2 \\ T_{ic} - T_{oc} = \frac{Q}{2h_{clad}} R_f \end{matrix}$$

$$T_{ic} - T_r = \frac{350}{2 \cdot 6} \cdot 0.45 = 13.125K$$

$$h_{clad} = \frac{0.18}{0.03 \cdot 0.06} = 6 \quad T_{ic} = 618.75K$$

$$618.75K - 13.125 = T_r$$

$$T_r = 605.625K$$



$$r=0.1: \text{ using } T(r) = \frac{Q(R^2 - r^2)}{4k} + T_s$$

$$T(r) = \frac{350 \left[(0.04 \text{ W/cm} \cdot \text{K})^2 - (0.1)^2 \right]}{4 \cdot 0.04 \text{ W/cm} \cdot \text{K}} + T_s$$

$+ T_s \quad 864.8K$
 $\downarrow$

2)
cont.

$$T_s = 864.8$$

$$T(r=0.1) = 846.425 \text{ K}$$

3)

$$L = 3.6 \text{ m}$$

$$LHR^0 = 400 \text{ W/cm}$$

$$\gamma = 1.2 \text{ s}$$

$$LHR @ z = 2.1$$

$$T_{cool \text{ out}}$$

$$C_p = 4200$$

$$m = 0.1$$

$$T_{cool in} = 580$$

$$z_0 = 3.6$$
$$\frac{z_0}{2} = 1.8 \text{ m}$$

$$LHR(1.8) = LHR^0 \cos \left[\frac{\pi}{2\gamma} \left(\frac{z}{z_0} - 1 \right) \right]$$
$$= 400 \cos \left[\frac{\pi}{2 \cdot 1.2 \text{ s}} \left(\frac{2.1}{1.8} - 1 \right) \right]$$

$$LHR(1.8 \text{ m}) = 391.2 \text{ W/cm}$$

3 cont.)

$$T_{\text{cool}}(1.8\text{m}) = T_{\text{coolint}} + \frac{2\gamma}{\pi} \frac{2\pi h L R^6}{\rho C_p} \left\{ \sin \dots \right\}$$

$$T_{\text{cool}}(z) = 580\text{K} + \frac{2 \cdot 1.25}{\pi} \frac{(1.8 \times 100)\text{cm} \cdot 400\frac{\text{W}}{\text{cm}}}{0.1\frac{\text{kg}}{\text{cm}^3} \cdot 4200\frac{\text{J}}{\text{kg} \cdot \text{K}}}$$

$$\left\{ \sin \frac{\pi}{2 \cdot 1.25} \right\} + \sin \left[\frac{\pi}{2 \cdot 1.25} \left(\frac{2.1\text{m}}{1.8\text{m}} - 1 \right) \right]$$

0.208

$$\{ \cdot 1.159 \}$$

$$T_{\text{cool}}(2.1\text{m}) = 716.42\text{K}$$

4) IC degrade @ ^{Crystal} bumps
S. I. FIMA

$$T_{avg} = 1200K$$
$$K_{ox} = \frac{1}{A + BT}$$

$$\frac{1}{B} \ln$$

$$A = 3.8 + 200 \times 0.05 = 13.8 \text{ W/cmK}$$

~~$A = 3.8 \times 200 \times$~~

$$B = 0.0217 \text{ cm/W}$$

$$K_{ox} = \frac{1}{13.8 + (0.0217)(1200K)} =$$

$$K_{ox} = 0.0251 \text{ W/cmK}$$

5) Assumptions needed for heat conduction equation to derive analytical form of temperature profile:

- 1) Steady State
- 2) axis symmetric, cylindrical
- 3) constant in z
- 4) constant k (thermal conductivity),
constant volume heat

6) In enrichment, UF_6 in gaseous form is used.

Enrichment processes take advantage

Of the mass difference between U^{235} and U^{238} , ~~chemical processes~~

and Kinetic Processes. Centrifuge-

based enrichment of U works

by using a centrifuge gas cylinder

that's rotated at high speeds.

It works because there's a

Strong centrifugal force that

causes the U^{238} to go to the

outside of the cylinder because of the

mass difference between U^{235} and

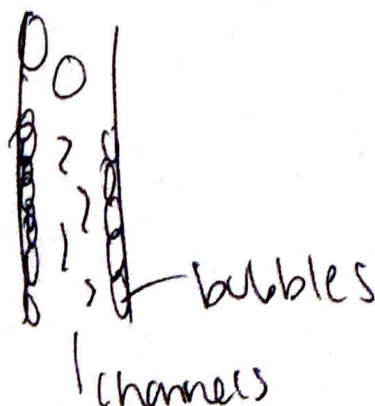
U^{238} . Therefore, U^{235} and U^{238}

can be separated/go into separate streams

7) Departure from nucleate boiling ratio (DNBR) is when ~~boiling~~ bulk liquid boils or regions of it boils and fluid travels in channels. Bubbles coat the solid surface (cladding) insulating the fluid moving through the channel.

This causes ~~the~~ the temperature of the surface to increase a lot.

$$DNBR = \frac{CHF}{q_{actual}} \quad \leftarrow \begin{array}{l} \text{critical} \\ \text{heat flux} \end{array}$$



7(cont.) Critical heat flux (CHFR)

is the flux when boiling becomes an ineffective form of transferring heat from a solid surface to liquid (CHFR).

8) Smear density is the ratio of volume of fuel to volume ~~of~~ of inner cladding ~~surface~~ surface.

$$V_{\text{fuel}} = \pi r_1^2 h$$

$$V_{\text{cladding inner}} = \pi r_2^2 h$$

$$\text{Smear Density} = \frac{V_{\text{fuel}}}{V_{\text{cladding inner}}} = \frac{\pi r_1^2 h}{\pi r_2^2 h} = \frac{r_1^2}{r_2^2}$$

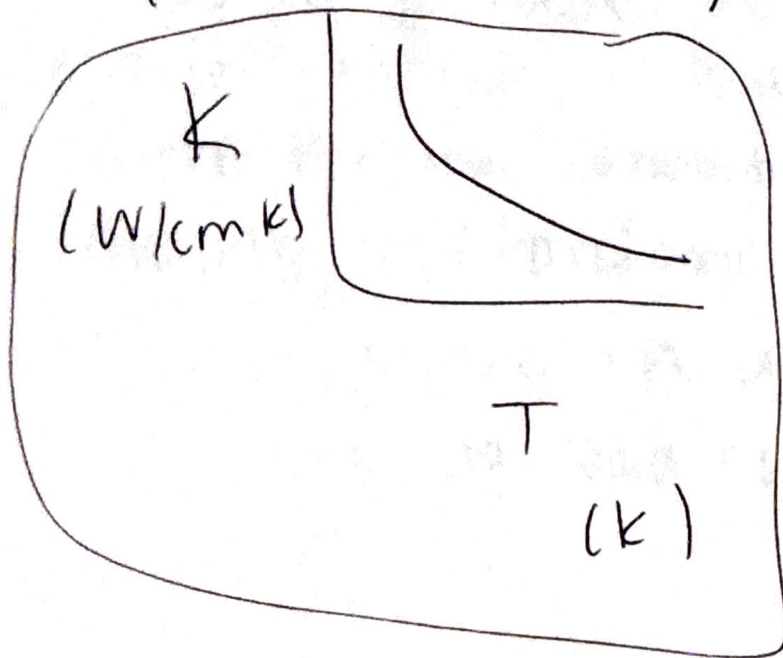
A reduced smear density is necessary for fuel fabrication to account for fuel swelling and cladding ~~pressure~~ shrinking on the fuel that can occur under irradiation/reactor conditions. Therefore, tight tolerances need to be used during fuel manufacturing.

a)

2 sources of thermal conductivity degradation in UO_2 are increasing temperatures and burnup. ~~these~~ This is shown

in $K_{ox} = \frac{1}{A+BT}$ where $A = 3.8 + 2.0WFI MA$.

FIMA relates to burnup and as fuel is burned, ~~there~~ fission products



act as obstacles increasing phonon scattering. Increase in temperature shown by T in

$K_{ox} = \frac{1}{A+BT}$ also increases phonon

scattering causes the ease of heat flow to suffer.

10) The role of cladding is to:

- retain fission products
- maintain fuel structure with room for coolant to flow
- Prevent corrosion of fuel by coolant
- keep coolant and fuel separate

Zr is a suitable cladding material because it has a low neutron cross section, high thermal conductivity, good mechanical quality, affordable, availability is good, stable in water. It's also resistant to void shelling.

11) Fuel system consists of:

- fuel
- gap
- cladding
- primary coolant

12) 3 aspects defined as constituting fuel performance are as follows:

- 1) How long the fuel performs without problems
- 2) How much heat transfers to coolant from fuel

~~3) Performance under accident conditions~~

- 3) How fuel performs under accident conditions

13) ~~3 ways~~ 3 ways space is discretized for numerical solutions are finite difference, finite volume, finite elements. Strengths and weaknesses are listed in a table below:

	Finite Difference	Finite Volume	Finite Element
Advantage/ Pro	• easy to implement • fast	• handles heterogeneous properties • conservative • handles different geometries	• handles ^{different} different geometries • handles different boundary conditions • heterogeneous properties • continuous
Weakness/ Con	• solves for specific points • limited by boundary conditions must be fixed T or dT/dr • can only handle rectangle geometry • can't handle heterogeneous properties well	• more complex than finite difference • limited ^{flux} boundary conditions; typically used in flow problems	• more complex • cost • more difficult to implement than finite difference

14) VO_2 FuelPositives:

- single phase fluorite up to melting point
- High melting point $\sim 2800^\circ\text{C}$
- relatively stable in water
- irradiation resistant
- no amorphization

Negatives:

- brittle (thermal stress fractures)
- poor thermal conductivity
- low linear heat rate

15) Conceptual difference between implicit and explicit time integration is forward Euler (explicit) determines future state of system ~~in~~ using the current state of the system to solve while backwards Euler (implicit) determines the future state of the system using the ~~current~~ future and current states of system to solve. Derivative difference. Explicit derivative is y'_n while implicit derivative is y'_{n+1} .